

DATONG, China

WMO: 534870

Lat: 40.0787N

Lon: 113.4116E

Elev: 3458

StdP: 12.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-2.3	-20.4	2.0	3.0	-16.9	2.5	5.7	20.8	18.1	18.8	18.6	4.7	340	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	89.6	63.9	86.8	63.0	84.3	62.9	70.7	80.5	69.2	78.8	67.7	77.2	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	117.1	74.9	66.3	110.1	74.0	64.7	104.0	73.0	37.0	80.6	35.7	79.0	34.3	77.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.5	17.7	15.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-12.3	93.8	3.5	3.4	-14.8	96.3	-16.8	98.3	-18.8	100.2	-21.4	102.7		
			-12.8	73.3	3.4	1.6	-15.2	74.5	-17.2	75.4	-19.1	76.3	-21.5	77.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	13.7	21.9	35.2	49.9	61.0	69.5	73.1	69.8	60.4	47.1	31.5	17.5
	DBStd	21.78	7.36	8.93	8.69	8.71	7.03	5.25	4.44	4.64	6.22	7.17	8.47	7.80
	HDD50	4195	1125	787	463	107	7	0	0	0	4	140	555	1007
	HDD65	7567	1590	1207	925	459	162	17	1	12	162	556	1004	1472
	CDD50	2734	0	0	4	103	347	586	716	613	315	50	0	0
	CDD65	629	0	0	0	4	37	153	252	160	23	0	0	0
	CDH74	5999	0	0	0	98	602	1590	2224	1227	253	5	0	0
	CDH80	1841	0	0	0	20	149	515	793	311	53	0	0	0
Wind	WSAvg	6.3	5.7	6.3	7.5	8.1	7.8	6.4	5.5	5.1	5.3	5.8	5.9	5.8
Precipitation	PrecAvg	14.8	0.1	0.2	0.4	0.7	1.1	1.7	4.0	3.5	1.9	0.8	0.3	0.1
	PrecMax	22.8	0.4	0.7	1.2	2.6	2.2	3.0	7.4	9.1	5.2	1.9	1.0	0.2
	PrecMin	8.4	0.0	0.0	0.0	0.0	0.1	0.4	0.9	0.7	0.3	0.0	0.0	0.0
	PrecStd	3.4	0.1	0.2	0.3	0.6	0.5	0.7	1.8	1.9	1.2	0.6	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	52.2	68.3	82.8	88.3	92.5	93.9	90.1	85.8	74.4	59.6	42.9
		MCWB	29.8	36.5	46.3	53.3	58.4	61.5	67.1	66.3	60.6	54.8	42.8	31.4
	2%	DB	35.0	45.7	61.0	76.2	84.1	88.6	90.2	86.1	80.1	69.2	54.3	37.7
		MCWB	26.3	33.5	42.0	51.0	56.4	60.5	65.6	66.0	59.6	51.2	40.1	28.7
	5%	DB	31.4	41.9	56.2	71.5	80.5	85.6	87.2	83.4	76.4	65.1	49.9	34.1
		MCWB	23.9	31.1	39.6	48.2	54.6	59.7	65.7	65.5	58.5	49.3	37.7	26.2
	10%	DB	27.8	37.8	51.5	67.0	77.0	82.7	84.5	80.9	73.2	61.2	45.6	31.0
		MCWB	21.5	28.8	37.1	46.6	53.6	59.6	65.7	65.0	57.1	47.8	36.0	24.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.3	38.1	47.0	56.2	61.6	67.3	73.0	72.9	66.5	56.5	45.0	32.3
		MCDB	39.7	48.9	63.7	74.0	78.3	78.9	85.1	81.2	76.1	68.9	54.3	41.3
	2%	WB	26.7	34.4	43.4	53.2	59.4	65.5	71.0	70.8	63.3	53.8	41.8	29.1
		MCDB	34.7	44.7	58.1	71.0	76.1	77.7	81.8	79.3	72.2	63.9	50.8	36.9
	5%	WB	24.1	31.6	40.7	50.5	57.4	64.2	69.8	69.2	61.0	51.9	39.3	26.7
		MCDB	30.8	40.4	54.4	67.5	74.1	76.5	80.4	77.4	71.5	61.4	48.6	33.3
	10%	WB	21.8	29.2	38.0	48.3	55.5	62.8	68.5	67.5	59.3	49.5	36.5	24.7
		MCDB	27.6	37.4	50.3	63.1	71.8	75.3	78.6	76.0	69.9	58.7	45.4	30.8
Mean Daily Temperature Range	5% DB	MDBR	21.7	22.9	23.4	24.5	24.2	22.3	19.4	19.4	20.8	20.9	20.6	20.3
		MCDBR	25.6	27.0	30.9	31.8	31.0	27.6	24.6	23.4	26.0	26.6	26.9	23.9
		MCWBR	18.8	17.6	17.2	14.6	12.4	8.9	7.8	8.6	10.2	13.0	15.7	17.0
	5% WB	MCDBR	24.3	24.7	28.2	27.3	25.0	20.6	18.8	18.1	19.9	22.1	23.3	22.3
		MCWBR	18.1	16.4	16.3	12.9	10.7	7.5	7.9	8.2	9.2	12.0	14.2	16.1
Clear-Sky Solar Irradiance	taub	0.323	0.389	0.416	0.471	0.514	0.590	0.569	0.564	0.501	0.439	0.353	0.327	
	taud	2.205	2.030	1.993	1.920	1.848	1.740	1.842	1.828	1.911	2.000	2.180	2.220	
	Ebn at Noon	260	254	262	255	247	228	232	228	233	233	245	247	
	Edh at Noon	32	44	51	59	64	72	65	64	55	45	32	29	
All-Sky Solar Radiation	RadAvg	838	1109	1522	1805	2012	1843	1752	1659	1416	1161	862	731	
	RadStd	71	75	127	106	143	142	88	116	91	80	102	67	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page